Fiscal Year:	FY 2026	**Task Last Updated:**	FY 09/29/2025
PI Name:	McGee-Lawrence, Meghan Ph.D.		
Project Title:	Osteocyte Plasma Membrane Disruptions in Skeletal Adaptation to Loading and Unloading		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	None		
PI Email:	mmcgeelawrence@augusta.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	706-446-0128
Organization Name:	Augusta University Research Institute, Inc.		
PI Address 1:	Department of Cell Biology and Anatomy		
PI Address 2:	1460 Laney Walker Blvd, CB1101		
PI Web Page:			
City:	Augusta	**State:**	GA
Zip Code:	30912	**Congressional District:**	12
Comments:			
Project Type:	Ground,New Investigation	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG2. App D: Flight and Ground Space Biology Research
Start Date:	12/01/2020	**End Date:**	11/30/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	4
No. of PhD Candidates:	4	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
Contact Email:	harry.jones@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/30/2025 per NSSC information (Ed., 11/1/24). NOTE: End date changed to 11/30/2024 per PI and NSSC information (Ed., 10/26/23).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hamrick, Mark Ph.D. (Augusta University Research Institute, Inc.) Johnson, Maribeth M.S. (Augusta University Research Institute, Inc.)		
Grant/Contract No.:	80NSSC21K0274		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The skeleton's ability to adapt to mechanical loading is crucial for bone health, as exercise promotes hypertrophy but disuse (such as from spaceflight) leads to bone loss. We were the first to report that small, transient plasma membrane disruptions (PMD) develop with in vitro and in vivo mechanical loading in bone osteocytes. These disruptions initiate skeletal mechanotransduction, suggesting PMD are stimuli recognized by osteocytes to regulate bone adaptation to its loading environment. Importantly, we consistently observe that ~20% of long bone osteocytes develop PMD with routine cage activity in mice, suggesting that formation of osteocyte PMD may be essential to bone's sensation of and response to normal gravitational loads. Accordingly, our central hypothesis is that osteocyte PMD formation is impaired during skeletal disuse, leading to bone loss. Our goals are to test the effects of disuse on osteocyte PMD formation, to determine whether osteocytes become sensitized to PMD formation with impaired PMD repair or survival during reloading, and to determine whether modulating osteocyte PMD formation and/or repair affect these processes. Our strategy is to test these concepts in an in vivo murine model of hindlimb unloading, as well as with in vitro osteocyte models of unloading (rotating wall vessel bioreactor) and reloading (fluid shear stress). Our goals align with the NASA Space Biology program as they target Research Topic 3 (Animal Biology Studies in support of Human Space Exploration)/ Sub-Topic AH1-E (Effects of fractional gravity provided by spaceflight centrifugation or ground microgravity/partial gravity analogs to gain insights into mechanisms of how animals sense, respond, and adapt to gravity shifts that are less than 1G) by discovering the contribution of osteocyte PMD formation (and hypothesized impairment during disuse) to the skeleton's adaptation to its loading environment. This project will yield a new understanding of how complex organisms adapt to the space environment, using a ground-based analog for disuse from spaceflight; we anticipate that derived data will advance strategies for skeletal maintenance and prevention of bone fractures during disuse to promote and support human space exploration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Disuse-induced bone loss, which occurs during prolonged exposure to microgravity during spaceflight and predisposes astronauts to risk of skeletal fractures, also occurs frequently on Earth in patients with spinal cord injuries, patients subjected to chronic/long-term bed rest, and in other cases of long-term decreased mobility. Furthermore, it is well understood that mechanical loading of the skeleton through physical exercise is beneficial for bone health across a wide spectrum of human patients, but there exists a substantial proportion of the population who cannot undertake regular vigorous exercise for a variety of reasons, including underlying health conditions, time constraints, or financial concerns. Therefore, understanding the fundamental mechanisms behind how bone senses and responds to changes in mechanical loading, and exploring ways to alter the skeletal response to a given level of mechanical loading (or withdrawal of loading), may lead to therapeutic interventions for improving bone health and reducing fracture risk.
Task Progress:	In the fifth and final year of this grant, with a no cost extension (NCE), we completed remaining work on the Specific Aims of the proposal. Major accomplishment were primarily in finishing studies with the disuse bioreactor conclusively establishing the role of disuse on PMD sensitization. These experiments are described below. We completed experiments inducing disuse in vitro using the Synthecon slow turning lateral vessel (rotating wall vessel) bioreactor system. Our data demonstrate that three days of culture in the bioreactor, which sufficiently up-regulates disuse-associated genes like sclerostin, sensitizes osteocytes to the formation of PMD during reloading; as significantly more PMD are formed with application of mechanical loading to disuse-exposed cells, as compared to application of loading to static control cultures. Moreover, we found that these same disuse conditions altered calcium signaling upon reloading; this was ascertained using a novel model we developed that permitted exposing disuse-cultured scaffolds (or static controls) to fluid shear to simulate reloading while performing real-time imaging of calcium signaling intensity. We submitted a peer-reviewed manuscript on these findings and received a decision of 'minor revisions'; we are completing revisions on this manuscript and anticipate resubmission prior to the end of the grant funding period. We continue to support the professional development of students, having supported a total of one postdoctoral fellow, four PhD students, five medical students, three undergraduate students, and a high school student during the completion of our funded experiments. Our final PhD student defended and successfully graduated in January 2024, and our postdoctoral fellow was promoted to 'Assistant Research Scientist' (a junior faculty position) in 2024. All of the trainees involved in the completion of this work have received authorship on either journal manuscripts or conference abstracts stemming from their contributions. Therefore, this grant has extensively supported the career development of the next generation of scientists.
Bibliography Type:	Description: (Last Updated: 11/12/2025)
Abstracts for Journals and Proceedings	Yu K, Tuladhar A, Dai C, McGee-Lawrence ME. "Integration of 3D osteocyte culture, microgravity simulation, and fluid flow reveals mechanisms of osteocyte mechanosensitivity and calcium signaling altered by disuse." ASBMR 2025 (Annual Meeting of the American Society for Bone and Mineral Research), Seattle, WA, September 5-8, 2025. Poster #Sun-157. Abstracts. ASBMR 2025 (Annual Meeting of the American Society for Bone and Mineral Research), Seattle, WA, September 5-8, 2025. , Sep-2025
Articles in Peer-reviewed Journals	Yu K, Tuladhar A, Dankberg S, Dai C, McGee-Lawrence M. "Integrating 3D osteocyte culture, microgravity simulation, and fluid flow reveals mechanisms of osteocyte mechanosensation and calcium signaling altered by disuse." Biomolecules. 2025 Oct 31;15(11):1534. https://doi.org/10.3390/biom15111534 , Oct-2025